ACE
Engineering College

An Autonomous Institution

Ankushapur (V), Ghatkesar (M), Medchal Dist. – 501 301

DEPARTMENT OF CSE(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Advanced AI-Powered Python Error Analyzer and Debugging Assistant

UNDER THE GUIDANCE OF
Mr.B.Avinash

PROJECT COORDINATOR
Mr.R.Rajesh

PRESENTED BY:
Anna Harish - 23AG1A6603
Banne Anand - 23AGIA6609
D. Swetha - 24AG5A6604

PROBLEM STATEMENT:

Python is widely used by beginners, but understanding and fixing errors remains a major challenge because python error messages are often technical and difficult to interpret. New programmers spend a lot of time searching for solutions without clearly understanding the cause of the error, which slows down learning and reduces confidence. Existing debugging tools mainly identify errors but do not explain them in a simple, human-friendly way. Therefore, there is a need for an intelligent system that can analyze python code and error messages and provide clear explanations and helpful debugging suggestions to assist beginners in learning python more effectively.

ABSTRACT

Python is one of the most popular programming languages because it is easy to learn and widely used. However, beginners often find it difficult to understand Python error messages and fix mistakes in their code. Most error messages are technical and do not clearly explain what went wrong or how to correct it, which makes learning slower and more confusing.

This platform introduces an **Advanced AI-Powered Python Error Analyzer and Debugging Assistant** that helps users understand errors in a simple and clear way. The system analyzes Python code and error messages using AI, Machine Learning, and Natural Language Processing to find the cause of errors and suggest easy solutions with good coding practices. A simple web interface makes the tool easy to use for beginners, helping them reduce debugging time and improve their programming skills.

Keywords:
Artificial Intelligence,
Python Debugging,
Error Analysis,
Machine Learning,
Natural Language Processsing

Current challenges faced

•**Complex Error Messages:** Python error messages are often technical and difficult for beginners to understand, making debugging confusing and time-consuming.

•**Lack of Clear Guidance:** Traditional debuggers only show where the error occurred but do not clearly explain why it happened or how to fix it.

•**Time-Consuming Debugging:** Beginners spend a lot of time searching online or trying multiple solutions to fix simple errors, which slows down learning.

•**Limited Learning Support:** Existing tools focus more on error detection rather than teaching users how to avoid similar mistakes in the future.

•**Dependency on External Help:** Many learners rely on forums, tutorials, or experienced developers to resolve errors, which is not always immediately available.

Literature Review :

Paper 1

Title: Automated Program Debugging Using Machine Learning
Authors: M. Monperrus, M. Martinez
Journal: IEEE Transactions on Software Engineering
Year: 2018
DOI: 10.1109/TSE.2017.2767601
Problem Defined:
Traditional debugging techniques require manual effort and deep programming knowledge. Beginners struggle to identify root causes of program errors efficiently.
Method / Approach:
The paper proposes machine learning models that learn from historical bug-fix data to automatically suggest fixes for software defects.
Key Findings:
ML-based debugging reduces developer effort and improves accuracy in identifying bug patterns.
Research Gap Identified:
Lack of beginner-friendly explanations and limited support for dynamic runtime errors.

Paper 2

Title: Intelligent Tutoring Systems for Programming Education

Authors: K. VanLehn

Journal: International Journal of Artificial Intelligence in Education

Year: 2016

DOI: 10.1007/s40593-016-0107-2

Problem Defined:
Students face difficulties understanding programming errors due to lack of personalized guidance.

Method / Approach:
Uses AI-driven tutoring systems to provide adaptive feedback based on learner behavior.

Key Findings:
Personalized explanations significantly improve student learning outcomes.

Research Gap Identified:
Limited integration with real-time Python debugging environments.

Paper 3

Title: Natural Language Processing for Software Error Explanation

Authors: S. Campbell, T. Nguyen

Journal: ACM Computing Surveys

Year: 2019

DOI: 10.1145/3317550

Problem Defined: Error messages generated by compilers are technical and hard to understand for beginners.

Method / Approach: Applies NLP techniques to translate technical error messages into simple natural language explanations.

Key Findings: NLP-based explanations improve comprehension and debugging speed.

Research Gap Identified:
Does not provide corrective code suggestions or learning feedback.

PAPER 4

Title: Deep Learning-Based Bug Detection in Python Programs

Authors: Y. Li, Z. Wang

Journal: IEEE Access
Year: 2020
DOI: 10.1109/ACCESS.2020.2978234

Problem Defined:
Manual bug detection in Python programs is time-consuming and error-prone.

Method / Approach:
Uses deep neural networks to learn syntactic and semantic patterns in Python code.

Key Findings:
Deep learning models outperform rule-based systems in bug detection.

Research Gap Identified:
Lacks explanation module for beginners.

Paper 5

Title: Program Repair Using Genetic Algorithms

Authors: W. Weimer, T. Nguyen

Journal: IEEE Software

Year: 2017
DOI: 10.1109/MS.2017.2650819

Problem Defined:
Fixing bugs manually requires significant human effort.

Method / Approach:
Applies genetic algorithms to automatically generate bug fixes.

Key Findings:
Automated repair can fix common bugs efficiently.

Research Gap Identified:
Not suitable for educational purposes due to lack of explanation.

Paper 6

Title: AI-Assisted Code Debugging Systems

Authors: J. Smith, R. Gupta

Journal: Springer – Software Engineering Notes

Year: 2021
DOI: 10.1007/s10664-021-09945-3

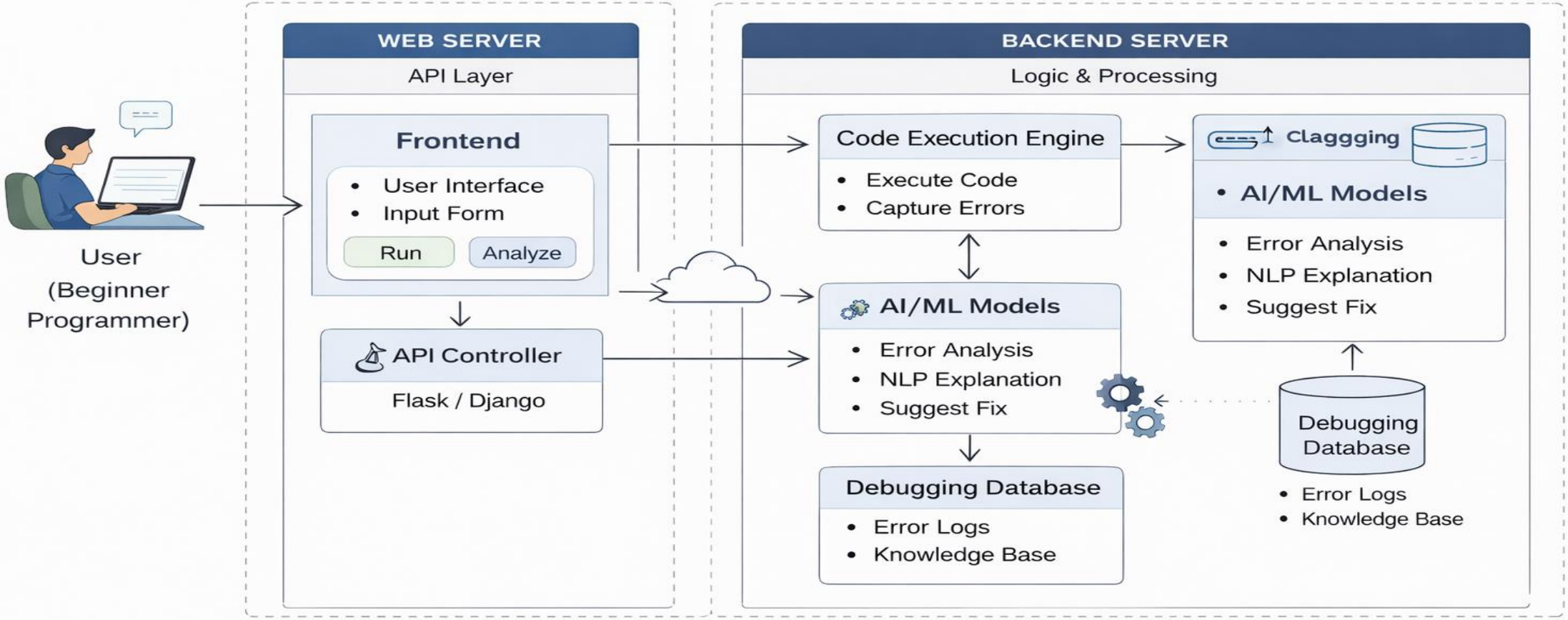
Problem Defined:
Existing debuggers focus on professionals, not beginners.

Method / Approach:
Introduces AI-based assistants that analyze code behavior and error patterns.

Key Findings:
AI assistants improve debugging efficiency.

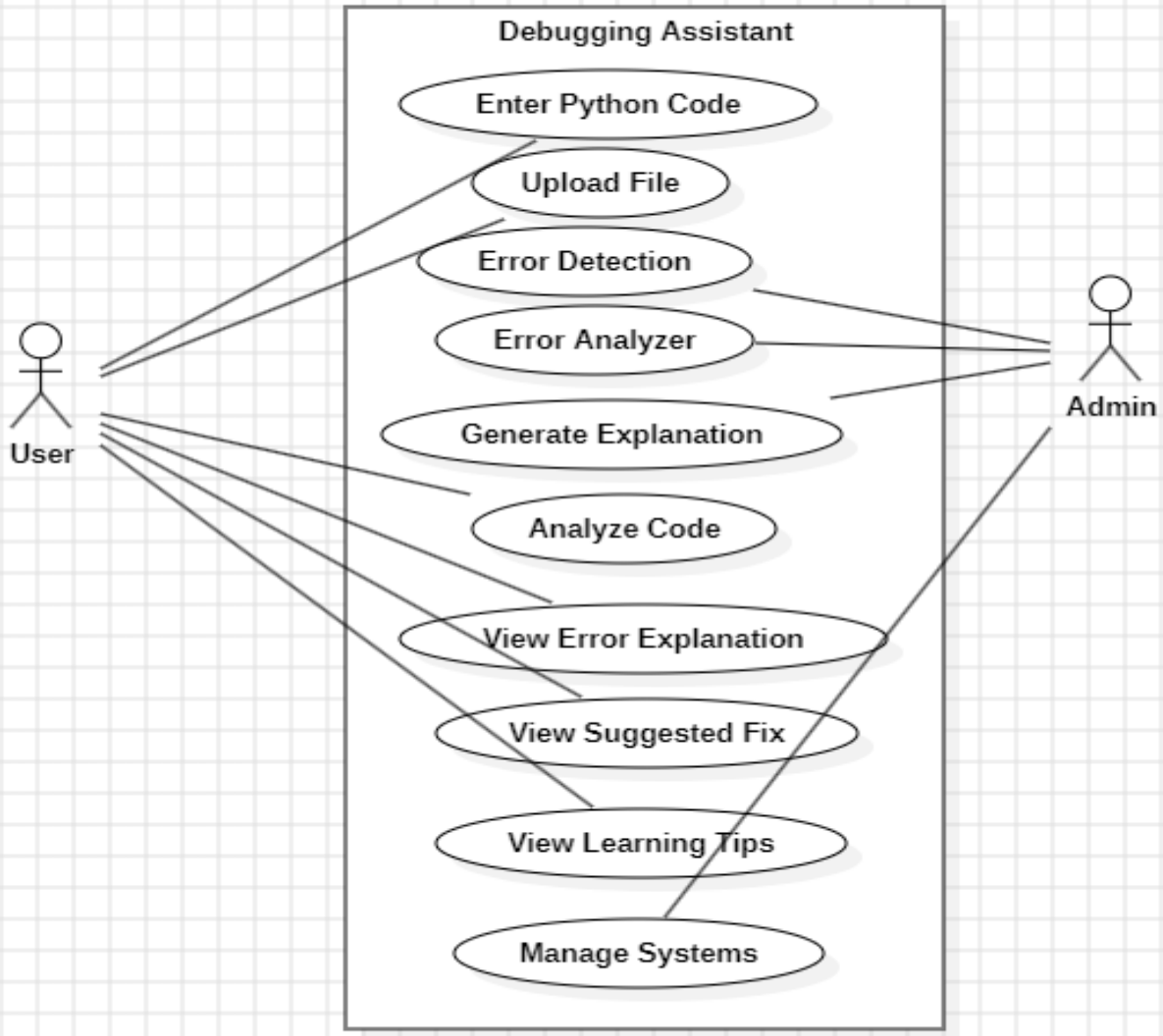
Research Gap Identified:
User interface complexity for novice programmers.

SYSTEM ARCHITECTURE:

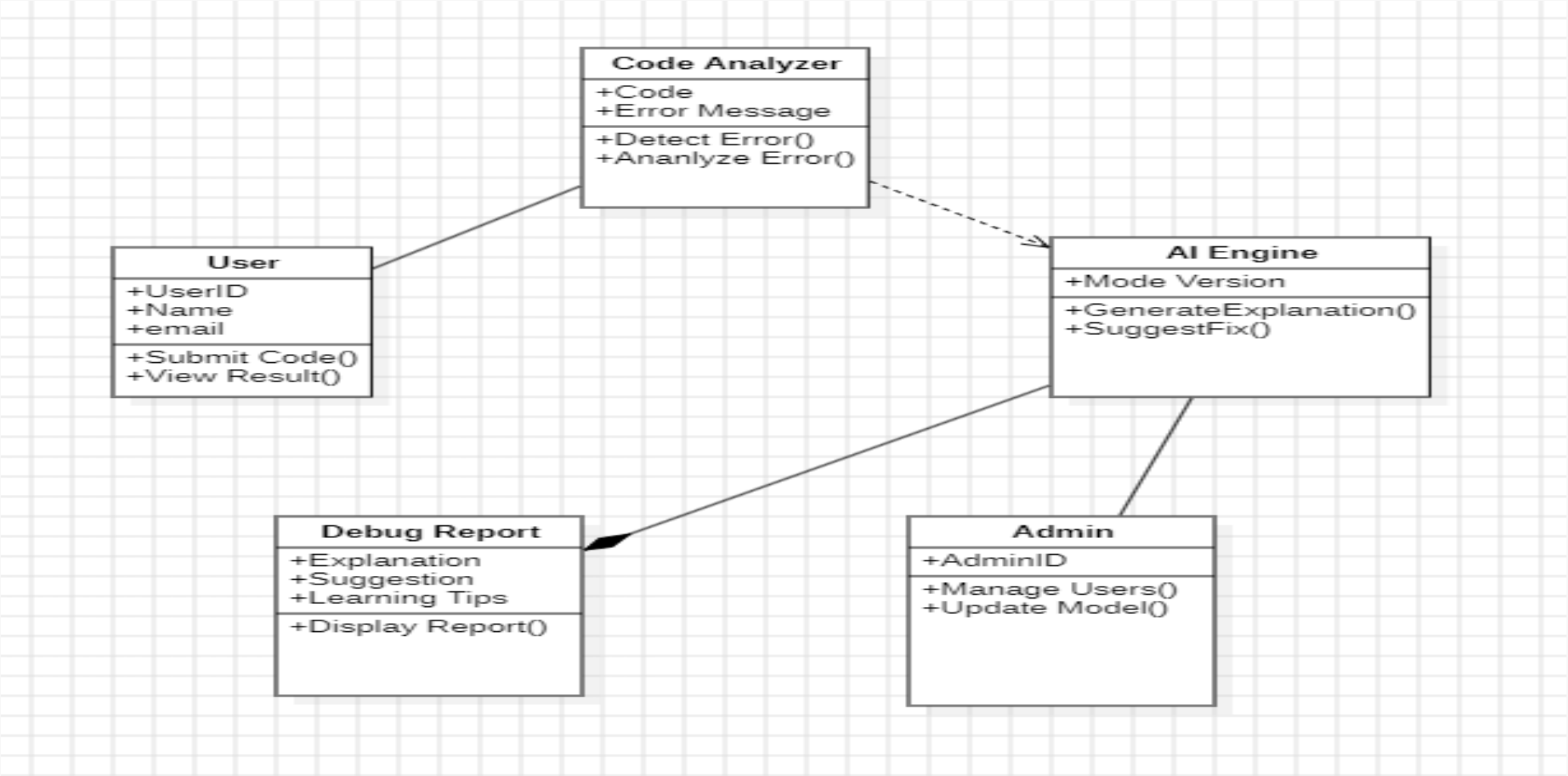


AI-Powered Python Error Analyzer and Debugging Assistant

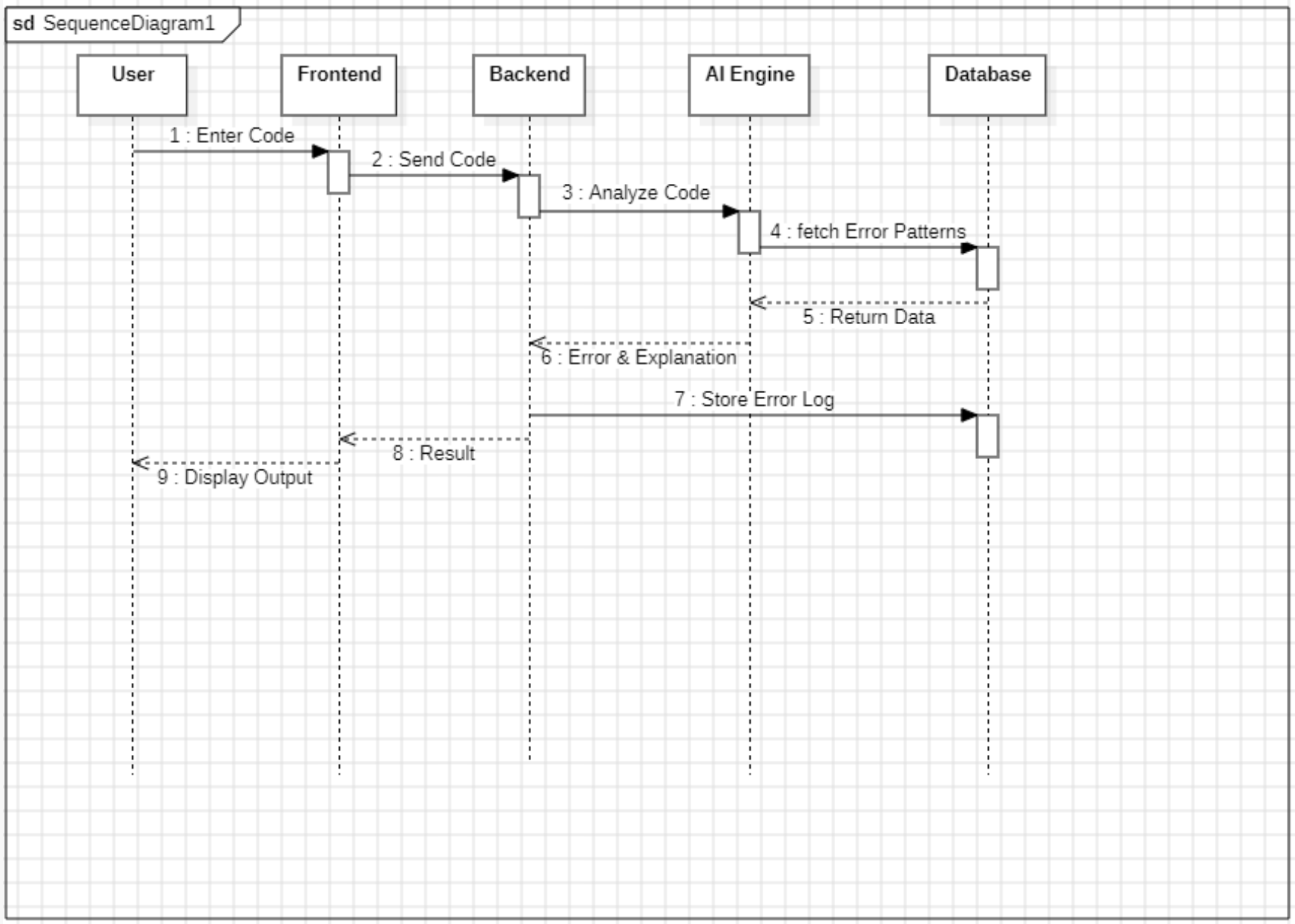
USECASE DIAGRAM



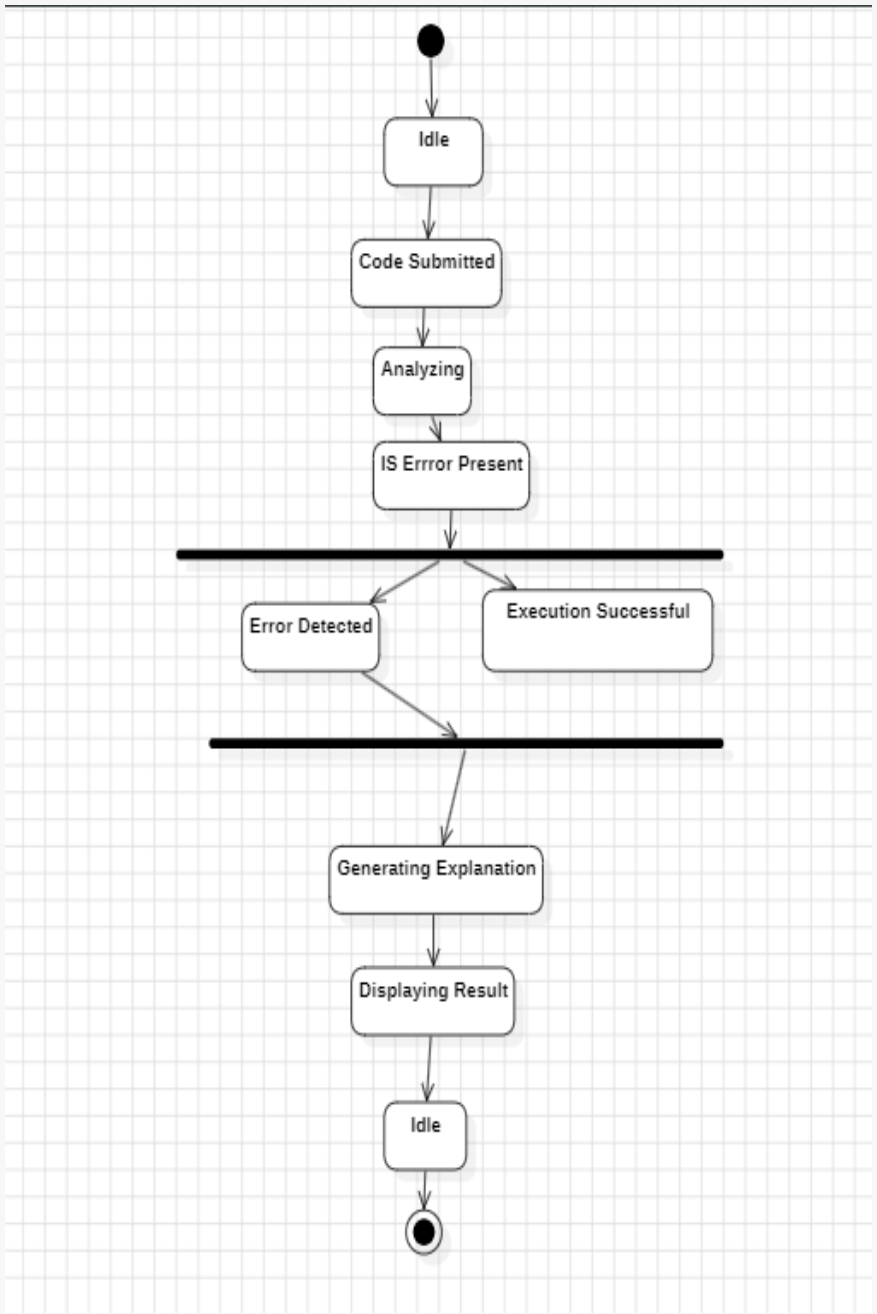
CLASS DIAGRAM



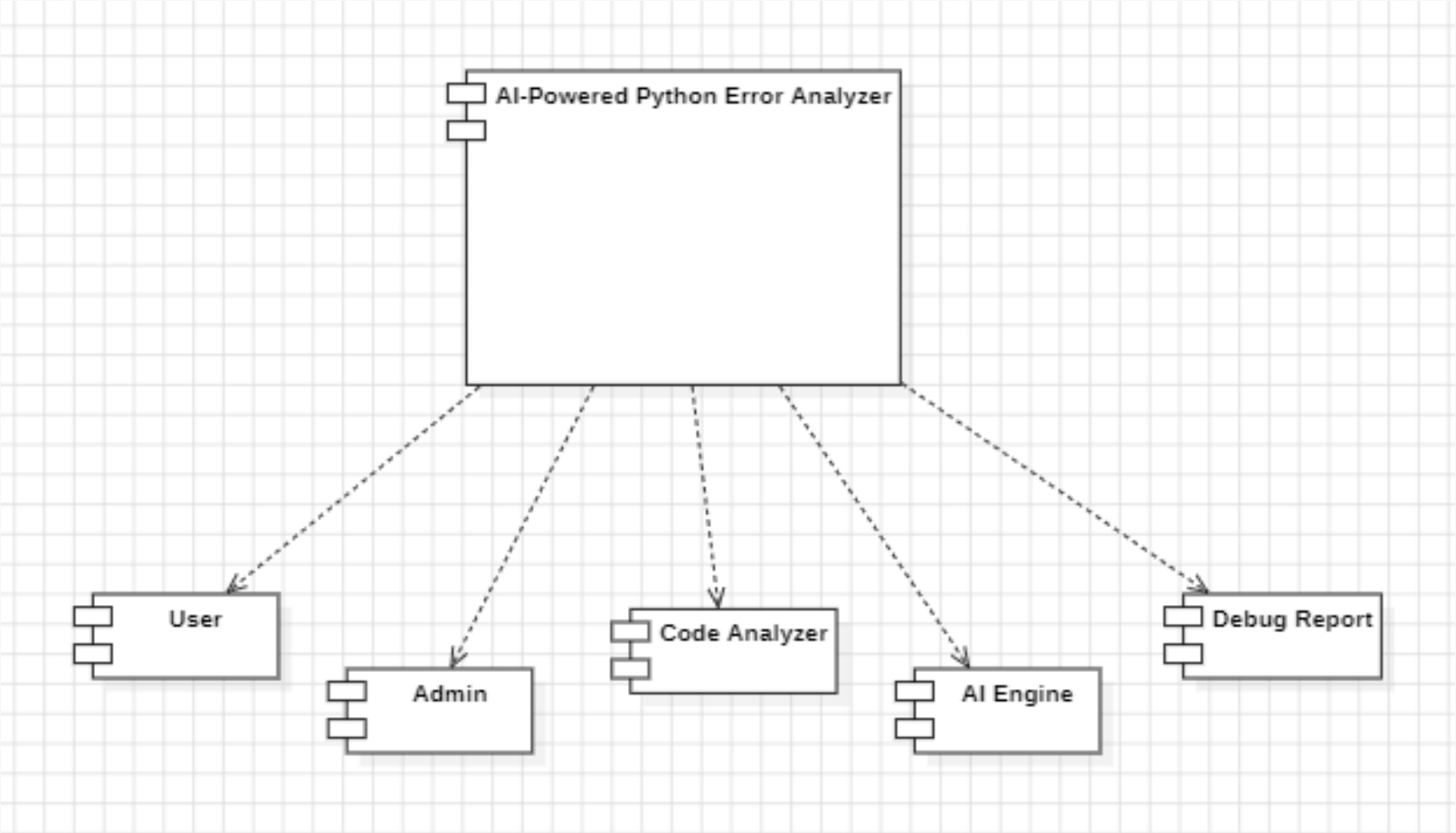
SEQUENCE DIAGRAM



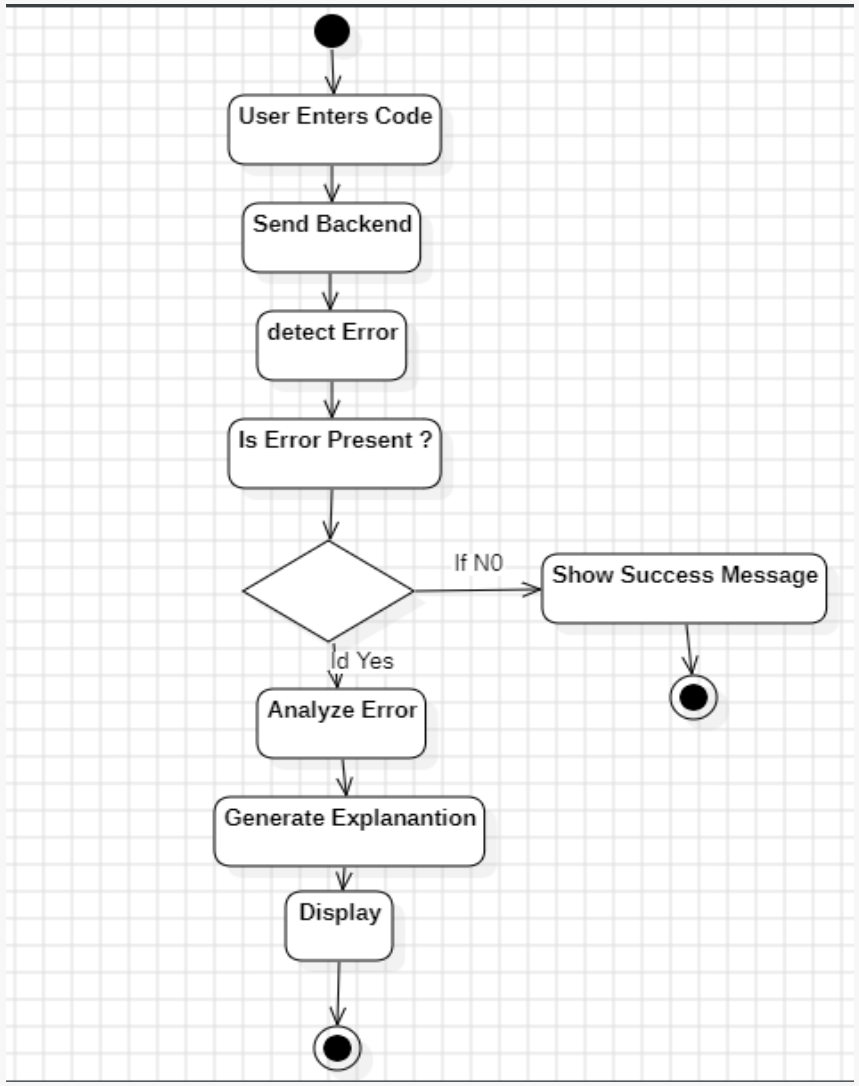
ACTIVITY DIAGRAM



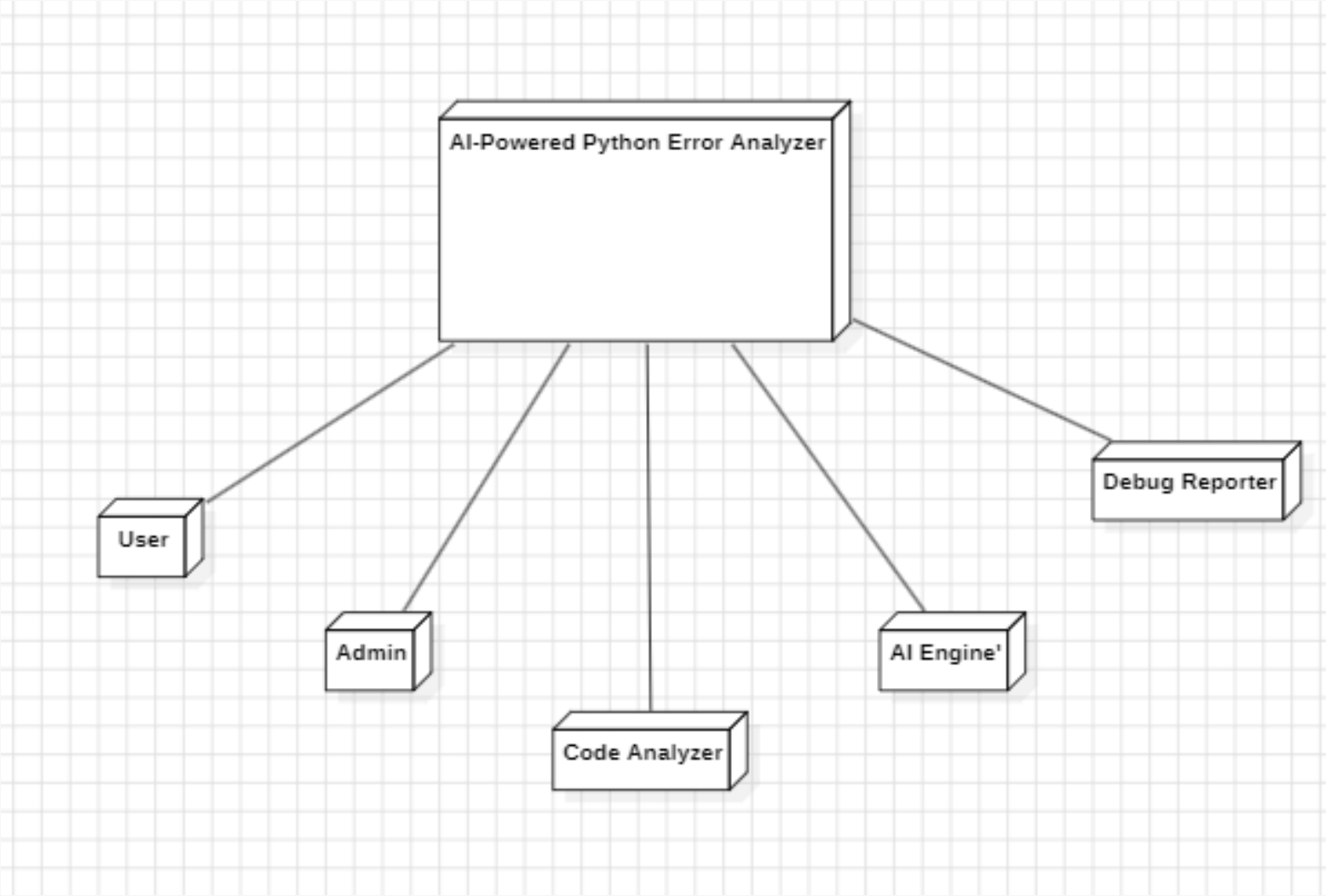
COMPONENT DIAGRAM



STATE CHART DIAGRAM



DEPLOYMENT DIAGRAM



OVERVIEW

The **Advanced AI-Powered Python Error Analyzer and Debugging Assistant** is designed to help beginners understand and fix Python programming errors easily. Many new programmers struggle with technical and confusing error messages, which slows down their learning process. This project solves that problem by using Artificial Intelligence, Machine Learning, and Natural Language Processing to analyze Python code and error messages. The system identifies the root cause of errors and provides simple, human-readable explanations along with suggested fixes and coding tips. It includes a user-friendly web interface where users can enter their code and receive instant feedback. By reducing debugging time and improving clarity, this tool helps beginners learn Python more effectively and confidently.

THANK YOU